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Preference-Based Assessments

Why Do Adults Value EQ-5D-Y-3L Health States Differently for Themselves Than for Children and Adolescents: A Think-Aloud Study



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ABSTRACT

Objectives: Following protocol, adults value EQ-5D-Y-3L health states from the perspective of a 10-year-old child. It remains unclear why adults value health states differently for themselves than for a 10-year-old child and whether the latter perspective is representative of adults' preferences for the 8 to 15 years age range of the EQ-5D-Y-3L. This study examines the reasons underlying (potential) differences in adults' health-state preferences for themselves, a 10-year-old child, and 15-year-old adolescent.

Methods: We conducted semistructured interviews using a think-aloud protocol with 25 participants who performed valuation tasks in July 2020 to August 2020. Using the Framework Method, we developed 28 categories grouped under 5 themes that illustrated the differences emerging from the interviews.

Results: Participants (A) deemed it more straining to perform valuation tasks for a 10-year-old child and 15-year-old adolescent than for themselves, (B) had a stable self-image, but varied in whom they imagined as 10-year-old child and 15-year-old adolescent, (C) focused on different dimensions and levels for a 15-year-old adolescent than for a 10-year-old child and themselves, (D) had various thoughts about nonhealth-related factors that influenced their preferences, and (E) gave up relatively few life-years for a 10-year-old child and 15-year-old adolescent, also to avoid others bearing a grudge against them.

Conclusions: Our results indicate that differences in adults' health-state preferences for themselves and a 10-year-old child largely result from differences in thoughts about nonhealth-related factors. They further indicate that health-state preferences for a 10-year-old child may not be representative of such preferences for the full EQ-5D-Y-3L age range.

Keywords: adolescents, children, EQ-5D-Y, health-related quality of life, health-state valuation, quality-adjusted life-year, qualitative methods, think aloud.

VALUE HEALTH. 2022; 25(7):1174–1184

Introduction

Economic evaluations of new health technologies for child and adolescent patient populations are increasingly applied.¹ The EQ-5D-Y-3L instrument was developed to measure health-related quality of life (HRQOL) in children and adolescents aged 8 to 15 years and calculate quality-adjusted life-years (QALYs) in economic evaluations (see [Box 1](#)).^{2–6}

The EQ-5D-Y-3L valuation protocol requires the application of discrete choice experiment (DCE) and composite time trade-off (cTTO) valuation tasks in a sample of the general adult population.⁷ The DCE tasks are used to determine the relative importance of the dimensions and levels of health states and the cTTO tasks to anchor the derived values on a 0 “dead” to 1 “full health” scale, which is necessary for calculating QALY gains in economic evaluations.^{5,7} The protocol further requires that adults perform the valuation tasks while “considering their views for a 10-year-old child,” which differs from the EQ-5D-3L and EQ-5D-5L valuation

protocols that require that adults perform the tasks for themselves while “considering their own health.”^{8,9} The perspective of a 10-year-old child was somewhat arbitrarily set; nevertheless, it was argued that there was no evidence to suggest that adults' health-state preferences for a 10-year-old child would be different from their preferences for a child or adolescent of a different age.⁷ Due to limited evidence and the likelihood that health-state preferences for children and adolescents would be age dependent, further research on this topic was recommended.⁷

The results of comparative research and the first EQ-5D-Y-3L valuation studies indicate that adults value similar health states differently for a 10-year-old child than for themselves.^{10–14} Indeed, health states are generally valued higher and within a narrower value range for children (see Lipman et al¹⁵ for a discussion on the implications for QALY calculations). Previous studies examining the reasons underlying adults' health-state preferences for themselves have shown that thoughts and beliefs relating to health and to nonhealth-related factors (eg, relating to support

BOX 1. EQ-5D-Y-3L instrument.

The descriptive system of the EQ-5D-Y-3L was developed in likeness of the EQ-5D-3L and comprises 5 dimensions: walking about, looking after myself, usual activities, pain or discomfort, and feeling worried, sad, or unhappy.^{3,5} In each of the dimensions, self-reported problems are expressed on a scale with 3 severity levels: no problems, some problems, and a lot of problems. This results in 243 (3⁵) unique 5-digit EQ-5D-Y-3L health states. The best (ie, least severe) health state described by the instrument is coded as 11111, describing “no problems” in each of the dimensions. The worst (ie, most severe) health state is coded as 33333, describing “a lot of problems” in each of the dimensions.⁵ Currently, the EQ-5D-Y-5L is being developed in likeness to the EQ-5D-5L.⁶

systems and life goals¹⁶) influence health-state values obtained from DCE and cTTO tasks. Insight into the reasons underlying observed differences between adults' health-state preferences for themselves and a 10-year-old child is currently limited to 1 study.¹⁷ Moreover, insight into the reasons that could potentially underlie any observed differences between adults' preferences for a 10-year-old child and a child or adolescent of a different age is lacking.

Therefore, the aim of this study was twofold: (1) to examine why adults value health states differently for themselves than for a 10-year-old child and the role of nonhealth-related factors in this context and (2) to examine whether and, if so, why adults value health states differently for a 10-year-old child than for a child or adolescent of a different age. To meet the latter aim, we asked adults to perform valuation tasks for a 15-year-old adolescent given that this age differed the most from 10 years, within the 8 to 15 years age range of the EQ-5D-Y-3L.⁵

Methods

Sample and Data Collection

We conducted semistructured interviews using a think-aloud protocol with 25 participants in July and August 2020. The participants were recruited by an independent research agency in The Netherlands and purposively sampled from their panel by age (18+ years), sex, health status, and having children (of any age) because we anticipated that these characteristics could influence participants' valuation of EQ-5D-Y-3L health states.¹⁸ We further anticipated that 25 interviews would suffice for reaching data saturation based on the sample sizes of comparable studies (eg, Dewilde et al¹³ and Karimi et al¹⁶).

The interviews were conducted face to face by the first author, who was trained by the EuroQol Group using the EuroQol Portable Valuation Technology (EQ-PVT) tool (version 1.8, adapted for this study).¹⁹ A total of 4 pilot interviews were conducted in a convenience sample to familiarize the first author with the interview and think-aloud protocols and to assess the feasibility of performing 8 DCE and 3 cTTO valuation tasks for each of the 3 perspectives (ie, adult, 10-year-old child, and 15-year-old adolescent). Based on the pilot results, we reduced the number of DCE tasks from 8 to 5, but kept the number of cTTO tasks to 3, per perspective to reduce participant burden and ensure sufficient time for follow-up questions.

Before conducting the interviews, the study objective was explained to participants and their written consent was obtained for audio recording the interviews and using their anonymized data for research purposes. Participants received €35 compensation for their time and effort, and ethical approval was obtained from the internal review board of Erasmus School of Health Policy & Management (institutional review board 20-02 Reckers-Droog, February 21, 2020).

Interview Protocol

The interviews were standardized using a protocol based on the EQ-5D-Y-3L valuation protocol,⁷ interviewer instructions for EQ-5D-Y valuation studies using EQ-PVT,¹⁹ and criteria for cTTO protocol compliance (see [Supplemental Material 1](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2021.12.014>).²⁰ Before starting the interview, the interview and think-aloud procedures were explained to participants, and it was explained that they did not have to verbalize thoughts or answer questions that they felt uncomfortable with. Participants were further asked to complete a short questionnaire on their socio-demographic characteristics, the EQ-5D-Y-3L (self-complete version on article), and a think-aloud practice task.^{5,21}

The interview commenced with an explanation of the valuation tasks to participants, after which they completed 5 DCE tasks and 3 cTTO tasks for themselves. The health states included in the tasks are presented in [Table 1](#). We selected health states from Kreimeier et al¹⁰ for the DCE tasks and from Ramos-Goñi et al⁷ for the cTTO tasks with the aim to examine participants' thoughts while trading-off between different dimensions and levels and between life years and HRQOL across the health-state severity range (see [Box 1](#)).

Participants then performed the DCE and cTTO tasks considering their views for a 10-year-old child and 15-year-old adolescent. The perspectives and health states, and their left-right position in the DCE tasks were randomized. On completion of the valuation tasks, a semistructured interview followed to obtain in-depth insight into participants' thoughts and beliefs relating to nonhealth-related factors and their influence on participants' health-state preferences ([Supplemental Material 1](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2021.12.014> includes the topic guide).

Table 1. EQ-5D-Y-3L health states.

Valuation task	Health states*
DCE [‡]	11 332 vs 22 222
	13 213 vs 32 331
	11 113 vs 11 121
	31 231 vs 32 313
	33 323 vs 21 133
cTTO [‡]	11 121
	22 223
	31 133

cTTO indicates composite time trade-off; DCE, discrete choice experiment.

*See [Box 1](#) for an explanation of the 5-digit EQ-5D-Y-3L health states.

[‡]Selected from Kreimeier et al.¹⁰

[‡]Selected from Ramos-Goñi et al.⁷

Table 2. Participant characteristics.

ID Characteristics									Interview duration [‡]
	Age	Sex	Education level*	Marital status	Children	Employment status	EQ-5D-Y-3L index [†]	EQ-VAS	
1	42	Female	High	Domestic partnership	No	Full time	1.00	94	67
2	62	Female	Medium	Married	Yes	Part time	0.81	80	74
3	58	Female	Medium	Divorced	No	Full time	1.00	95	84
4	61	Male	High	Single	No	Self-employed	1.00	95	78
5	34	Male	Medium	Married	Yes	Full time	0.81	68	80
6	42	Female	Medium	Married	Yes	Part time	0.84	75	72
7	59	Male	High	Married	Yes	Full time	0.72	75	97
8	37	Male	High	Single	No	Full time	0.84	77	91
9	70	Female	High	Single	No	Part time	0.84	75	85
10	42	Female	High	Married	No	Part time	0.81	70	74
11	39	Male	High	Single	Yes	Self-employed	0.84	74	80
12	28	Female	High	Single	No	Full time	1.00	80	86
13	72	Male	High	Domestic partnership	Yes	Retired	0.84	90	70
14	44	Female	Medium	Married	Yes	Unable to work	0.81	60	80
15	61	Male	Medium	Married	Yes	Part time	1.00	90	88
16	20	Female	Medium	Single	No	Student	0.84	75	72
17	24	Male	High	Single	No	Student	1.00	80	85
18	45	Female	High	Domestic partnership	Yes	Part time	1.00	92	78
19	22	Female	Medium	Single	No	Student	1.00	92	79
20	33	Male	Medium	Single	No	Part time	0.34	80	85
21	32	Male	High	Domestic partnership	No	Self-employed	1.00	83	77
22	31	Female	High	Single	No	Full time	0.25	80	73
23	48	Female	Medium	Single	Yes	Full time	0.72	70	84
24	24	Male	High	Single	No	Student	0.81	60	114
25	62	Male	Medium	Married	Yes	Retired	0.84	84	64

EQ-VAS indicates EuroQol visual analog scale; ID, identification. EQ-VAS range from 0 “Worst imaginable health” to 100 “Best imaginable health”.⁵

*Medium = middle vocational and secondary school; High = higher vocational and academic education.

[†]EQ-5D-Y-3L index value is calculated using the EQ-5D-3L value set for The Netherlands.²³

[‡]Interview duration presented in minutes.

Data Analysis

We analyzed the data using the Framework Method in the 5 stages described in [Box 2](#).

Results

The participant characteristics are presented in [Table 2](#)⁵ ([Supplemental Material 2](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2021.12.014> includes the summary statistics). Data saturation was reached after 21 interviews.

Based on the framework analysis, we developed 28 categories grouped under 5 themes ([Supplemental Materials 3](#) and [4](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2021.12.014> include descriptions of the themes and categories and

the DCE and cTTO results, respectively). A narrative summary of the framework is provided below, with [Table 3](#) presenting illustrative quotes (Q), along with participants' identification (ID).

Theme A: Performing the Valuation Tasks

Participants commented broadly on health states while performing the valuation tasks. They mentioned that dimensions were not always clear and that it was difficult to imagine or consider certain health states as being realistic (Q1). Participants also commented on the hypothetical context in which they stated their preferences. Some mentioned being willing to state preferences in this context, but being uncertain whether they would be or would have similar preferences, when asked in real life (Q2). Participants further commented on the underlying assumptions of the tasks. They struggled with the lack of possibility to change

Table 3. Framework with illustrative quotes.

Code	Theme	Category	#	Quote	ID
A	Performing the valuation tasks	Clarity and coherency of health states	1	It's always nice when someone is able to walk and not in any pain. But then to also have problems with washing and dressing ... I find that a difficult combination [for a 15-year-old adolescent] that I just can't imagine.	4
		Hypothetical context	2	Hypothetically, I am prepared to give up three life years [for a ten-year-old child], but whether I am in real life I don't know.	5
			3	You say there is nothing I can do to change this health state, but I keep thinking that I can.	11
		Maintaining focus	4	It's becoming harder and harder to think about this [for a 10-year-old child], because of all the misery that I read.	25
		Satisficing behavior	5	I feel like running away from it, from making decisions [for a 10-year-old child]. Go on ... I don't want to think about it any longer.	25
			6	When I performed the tasks for myself, I didn't realise that I would also be performing them for someone else. For myself, I find them quite easy but for someone else ... I just say 'no' to all of them.	20
			7	It's more difficult to choose based on my emotions than to make a rational choice [for a 15-year-old adolescent]. I'm more inclined to say 'take away a few life years,' because I feel more distant from it.	5
			8	Now [lives] A and B are equal because you can transfer the responsibility [of performing the task] to the person himself. This is it then, the 15-year-old adolescent is now 20, I now say: decide for yourself.	22
B	Perceptions about the person living in the health states	General image of the person living in health states	9	I always say 'everything but death' ... I am not the kind of person to give up easily.	6
			10	I've always said that I'm not the type that would want to suffer or be sick for a long time ... that's just not who I am.	11
			11	Very worried ... do you wish that upon him, her, him or her? It's best not to be worried [for a 15-year-old], but pain ... can they handle that?	6
		Imagining (someone else) living in health states	12	I'm trying to imagine what choice I'd make if I were a [10-year-old] child. I'm an adult, so choosing for someone else is not that hard, but if you relate it to yourself it becomes more difficult. I make this choice solely for a child now, I made the previous choice much more from my own perspective.	13
			13	I tried to recall my own memories and imagine some classmates from the past. I couldn't imagine my son as a 15-year old, so I made the choices with a generic image of an adolescent in my mind.	5
		Relationship to the person living in health states	14	This is going to be tough because I'm seeing my own son in front of me. It's very personal, but this will help me to give better answers [for a 10-year-old child]... I think.	5
			15	I find it more difficult to imagine a 10-year-old child than someone at the age of 15. I don't have children and I can imagine myself much better as an adolescent than when I was 10... that age feels more distant to me.	
			16	It's a bit of a generalisation, but I don't think my choices would have been very different if I had thought about a boy instead of the girls I had in mind.	19
		Family and peer relationships	17	A 10-year-old child and 15-year-old adolescent both go to school, want to go outside, and do their day-to-day things. But an adolescent is becoming independent, and a child is much more tied to home. Their needs are the same, but it's still different.	6
			18	Some problems can ultimately become a lot of problems [for a 15-year-old adolescent]. If you can't participate in daily activities ... adolescents always live in groups, if you deviate, you're bound to be lonely.	10
		Cognitive and developmental aspects	19	They'd both find it hard if they couldn't move that well. But a 15-year-old might understand it better, how heavy that weighs. Maybe I'm saying something really wrong here, but I think that because they have more life experience ... if they have to miss it, have to give it up ... it would be much harder [than for a 10-year-old child].	6

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Table 3. Continued

Code	Theme	Category	#	Quote	ID
		Resilience and coping skills	20	You'll have problems with the fact that you're unable to do it yourself. Children already receive help; their parents wash and dress them anyway. In that regard, there is more of a tendency to say ... they still can't do it. Whereas I would again not be able to do it. I think it would be the same for a 15-year-old as it is for me.	16
C	Interpreting and assessing the health states	Experience with health states	21	I don't like either health state, but I choose A because then you'll have some problems with everything. That could be my prospect, I have an 81-year-old sister who is familiar with that [health state]. You can't stop ageing ... I choose A, but I'd rather stay healthy until I'm 90.	9
			22	I had co-workers with daughters aged 12 and 18. One had cancer and the other got hit by a car. I have witnessed firsthand how much impact that has... that came to mind.	4
			23	I'm just thinking about how important it is to be able to walk in life. At one point, my grandmother couldn't walk anymore ... that caused many more problems than just with walking.	21
			24	If a 10-year-old child stays that way for 10 years, then you're 20. I can't imagine that those first ten years have been very pleasant... so it's a bit like a life in sadness. You've lived the first 20 years of your life like this, and you've had no alternatives.	9
		Dimensions and levels of health states	25	I think it is more important for a 15-year-old to be physically mobile than for a 10-year old. A 10-year old still needs his parents for getting washed and dressed, so problems with that would actually not be a problem. But a 15-year old must be able to function independently.	8
			26	I think that if your body starts to develop [in puberty], it isn't nice if someone has to help you with washing yourself. I could perhaps get over that, but I don't think that would be nice for someone at the age of 15.	10
		Duration of health states	27	I didn't take the happiness [of a 10-year-old child] into account, because ... what I can remember about myself from when I was a child was that something could happen and I would be super sad about it and five minutes later, I had already forgotten about it. At the age of 15, that was very different.	20
			28	When you're in puberty, you're always worried, sad, or unhappy ... you have your ups and downs, which just fits adolescent behaviour. I think it's more acceptable because you're just in that stage of life.	12
		Consequences of health states for persons not in health states	29	If I choose [life] A, I'm going to die and if I choose [life] B, I'll be in pain. Pain can really cloud your mind and then you'll be a burden to someone else ... so, I'll choose [life] A.	2
		Consequences of health states for person in health states	30	There's a lot of bullying during puberty and a 15-year-old can easily be ostracised. I think 10-year-olds are more caring ... my gut says it will be harder for an adolescent.	11
D	Influencing factors	Housing and available facilities	31	I think it matters where you live, what area you live in. We moved to a village where lots of people know each other and everything to meet our daily needs is close by. And I don't have to leave the village when I need medical care.	7
		Support system	32	As an adult, you try to protect a 10-year-old as much as possible... make sure they can go and do whatever they want. That's how I look at it, also for a 15-year-old, but in relation to a 15-year-old you know there are other influences at play.	15
		Normal and acceptable health	33	A [10-year-old] child should not be worried, should not be sad or unhappy ... so I choose [health state] B.	14
			34	I have always been told and taught that it is more tragic when parents lose their children... so that weighs a lot heavier than children losing their parents.	24
		Preferences of the person living in health states	35	I think you are a bit more careful because you're ... it's kind of hard to say what someone else would consider worse. Maybe they would say that walking is important.	16
			36	I think ... if you'd ask a [10-year-old] child they'd say they rather become 20 with a lot of pain, than how I look at it now. They'd give a different answer out of innocence and their will to live.	1

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Table 3. Continued

Code	Theme	Category	#	Quote	ID
		Preferences of persons not living in health states	37	I might want to live longer myself, but it doesn't do my children any good when I am lying sick in my bed for 10 years. So I took that into account ... a little bit.	11
		(Not) having children	38	This is a theoretical choice of course ... but I almost know for certain that if I had a 10-year-old child, I would have chosen life B.	21
		Life events and goals	39	Yes, now [lives] A and B are the same. The [10-year-old] child is now ... that is a nice age. They finished secondary school healthily and had their youth.	4
		Hope	40	[For a 15-year-old adolescent] one year can be the difference between falling in love for the first time and being able to ride a scooter. So I choose [life] B, I just hope that he is surrounded by lovely people who can help him and alleviate his worries.	5
			41	My husband has a nephew who is in a wheelchair, he has had a muscle disease since birth. The doctors had already given up on him, but he turned 25 last week. So you see ... when you give a child that much love ... If he dies now, after so many years ... he is an adult.	14
			42	Where that turning point lies exactly is difficult to say, because medical advancements are not that precise either.	5
			43	I'm constantly seeing a [small] 10-year-old child in front of me ... you never know what will happen in those years, there might be some medical development that can reduce your pain. I just think that you should try that, so I choose life B.	22
E	Stating a preference	Stating a preference for someone else	44	Here we go again ... the [10-year-old] child is very worried, sad or unhappy ... again all these impairments. I don't want to bring that upon my child for ten years.	11
			45	I choose [life] A because I would hate to give [a 15-year-old adolescent] an unhappy life.	3
		Giving up life-years	46	A 10-year-old child can't choose for themselves. You don't ask a 10-year-old ... do you want to die now or in six months?	12
			47	Of course, a 10-year-old child is not yet able to make big decisions, for example, about euthanasia. Certainly not in light of their brain development, so I don't want to put that in his hands.	22
			48	My [6-year-old] child would be able to deal with these questions. It would be difficult and make him sad, but it would be better to make the choice together than to exclude him.	5
			49	I think a 15-year-old should choose for himself. They have a broader perspective and know how they feel about things. For a 10-year-old that's more difficult, so then it's more a matter of talking about it.	5
			50	To choose for these children is really very difficult, because I think that a 15-year-old can certainly ... but also a ten-year-old child, I would allow them to consider the matter. I find it difficult that I would be the doctor or the person who has to decide, I'd rather leave it up to them.	1
		Legitimacy of choice	51	I find it difficult to imagine what it's like to be depressed ... I notice that I'm downplaying it. I'm making this choice [for a 15-year-old adolescent], while I have a blind spot. Within my own moral framework, I am not entitled to make this choice, because ... who am I?	5
			52	So in your choices, you take into account what you think, but also what you think the child or the adolescent might think. So I'm kind of seeking for a compromise, for some middle ground. (ID 12)	12
		Consequences of choice	53	It's a lot easier when a [10-year-old] child can decide for himself and you can support him. Because you don't want a child to blame you ... to later resent your choice.	5
			54	You still want to keep your child with you, to protect him all those years. After 10 years, he will die at the age of 20, but you will at least have developed a bond with him so that he doesn't feel, sort of like ... my parents ditched me.	14
		Egoistic vs altruistic preferences	55	I think that a [10-year-old] child ... that the child would say ... let me go. But then you're in that position and you think ... you are my child... I am not letting you die [...] I would keep on choosing [life] B, I would not give up.	14

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Table 3. Continued

Code Theme	Category	# Quote	ID
		56 It's difficult because my own interests are coming first. It's a bit selfish ... I'm not listening to what the [10-year-old] child would like.	22

indicates number; ID, identification.

health states and to end lives prematurely and believed that these assumptions would not hold in reality (Q3).

Some participants mentioned that it was easier to perform the valuation tasks considering their views for a 10-year-old child and 15-year-old adolescent than to perform the tasks for themselves. The choice between health states had become more abstract and less straining because they had already stated health-state preferences for themselves several times. For most it was more difficult to perform the tasks considering their views for a 10-year-old child and 15-year-old adolescent. Some mentioned that it had become difficult to maintain focus and value health states (Q4).

Participants sometimes exhibited strategies to shortcut or make the valuation task easier for themselves (here referred to as heuristic valuation processes)^{24,25} by focusing solely on or actively discarding dimensions and levels—ostensibly making it easier for themselves to interpret and assess health states. Some exhibited speeding (ie, stating preferences quickly just to finish the tasks) and straight lining (ie, consistently being unwilling to give up life-years) behavior—ostensibly avoiding feelings of discomfort when stating preferences for a 10-year-old child and, to a lesser extent, 15-year-old adolescent (Q5; Q6). Other participants distanced themselves emotionally from the tasks or were willing to give up a few life-years, but only up to the point where they believed that the 10-year-old child or 15-year-old adolescent would be old enough to be handed over the responsibility of performing the task and attaching a more accurate value to the health states themselves (Q7; Q8).

Theme B: Perceptions About the Person Living in the Health States

Participants' self-image (ie, their mental picture of themselves) was stable between tasks (Q9; Q10). This was different for participants' image of a 10-year-old child and 15-year-old adolescent. They used a wide range of personal pronouns and of singular and plural nouns when referring to a 10-year-old child or 15-year-old adolescent. Participants were often ambiguous and uncertain about whom to imagine, and this varied greatly within a perspective and across tasks (Q11). They alternated between imagining their (grand)child, themselves as a 10-year-old child or 15-year-old adolescent, and an unknown 10-year-old child or 15-year-old adolescent (Q12; Q13). Related to this, participants had different perceptions about their relationship with the 10-year-old child and 15-year-old adolescent. Sometimes, they imagined someone to whom they felt emotionally attached, whereas at other times they imagined someone from whom they felt more emotionally detached (Q14). Participants who drew on their childhood memories while performing the tasks mentioned that it was easier for them to remember what it was like to be a 15-year-old adolescent than a 10-year-old child. The latter memories were more distant and vague (Q15).

Some mentioned that the characteristics and interests of a 10-year-old boy were different from those of a 10-year-old girl (eg, they might prefer different types of games). Nevertheless, they believed that these differences did not affect their preferences for

a 10-year-old child (Q16). Participants also mentioned that the characteristics and interests of a 10-year-old child and 15-year-old adolescent were different. Generally, a 10-year-old child was regarded as "very young," which was reflected in the frequent use of diminutives in relation to a 10-year-old child, and a 15-year-old adolescent as "coming of age." An important difference between a 10-year-old-child and 15-year-old adolescent concerned the relatively strong orientation of the former toward their family and of the latter toward their peers (Q17). Participants believed it was particularly important for a 15-year-old adolescent to be able to keep up with their peers. They believed that having problems, for example, with walking about or doing usual activities, would negatively affect the emotional and social development of a 15-year-old adolescent (Q18).

Participants believed that an adult and 15-year-old adolescent would be better able to understand and appreciate living in an (impaired) health state than a 10-year-old child. Consequently, it would also be more difficult for an adult and 15-year-old adolescent to live in such a health state (Q19). They believed that a 10-year-old child was more flexible and resilient and would have better coping skills than an adult and 15-year-old adolescent. A 10-year-old child would simply not yet know any better, considering that a 10-year-old has little health experience (be it with good or bad health) to compare health states with and will likely still receive help from their parents anyway (Q20).

Theme C: Interpreting and Assessing the Health States

Participants relied on their memories and experiences to interpret and assess health states. Memories were often about participants' own experiences with impaired health but included experiences of their (grand)children or parents, other family members, and coworkers (Q21; Q22).

Memories and experiences seemingly made it easier for participants to interpret and assess health states. Nevertheless, they sometimes led participants to read more into the health states than what was described (Q23; Q24). For participants without such memories or experiences, it was more difficult to interpret and assess health states. Some expressed concerns about not being able to properly value health states because of this.

The relative importance of dimensions and levels was dependent on the perspective. Participants generally believed that any problems they themselves or a 10-year-old child would have with walking about, looking after themselves, and doing their usual activities could be solved relatively easy with help from others. They believed that a 15-year-old adolescent would likely not want any help from others—particularly not with looking after themselves (Q25; Q26). Participants further believed that the same level of problems in the same dimension would not always affect them, a 10-year-old child, and 15-year-old adolescent the same way. For example, they believed that having pain or feeling worried, sad, or unhappy would have little impact on a 10-year-old child, because this was often short-lived (Q27). In turn, participants believed that feeling worried, sad, or unhappy would have a strong impact on a 15-year-old adolescent. Nonetheless,

they believed that problems in this dimension were relatively unimportant given that feeling worried, sad, or unhappy was only normal for someone going through puberty (Q28).

When interpreting and assessing health states, participants considered the possible consequences of living in a health state for their own lives and the lives of a 10-year-old child and 15-year-old adolescent. In particular, they considered how certain (in)visible impairments might affect their participation in everyday life and their relationships with others (Q29). Several participants expressed concerns about them being dependent on others and becoming a burden to their loved ones and about a 15-year-old adolescent being bullied by their peers (Q30).

Theme D: Influencing Factors

Participants made comments on a wide range of nonhealth-related factors that influenced their health-state preferences. We grouped these factors under the following categories: housing and available facilities, support system, normal and acceptable health, preferences of the person(s) not living in health states, (not) having children, life events and goals, and hope.

Suitable housing, available facilities, and having a (professional) support system mostly influenced participants' health-state preferences for themselves (Q31). Most participants believed it was only normal that a 10-year-old child and, to a lesser extent, 15-year-old adolescent would be supported by their parents, although they believed their support had only limited impact on the latter (Q32).

Participants' preferences were also influenced by notions of normal and acceptable health states. For themselves, such notions were often related to the acceptability of physical impairments at an older age. For a 10-year-old child and 15-year-old adolescent, such notions were arguably more normative. Participants generally believed that a 10-year-old child should be happy, whereas a 15-year-old adolescent should be, at least, a bit unhappy. They also believed that a 10-year-old child and 15-year-old adolescent should have a normal life span and outlive their parents (Q33; Q34). Participants further tried to imagine what preferences the 10-year-old child, 15-year-old adolescent, and others not living in the health states (eg, their family members) would have and incorporated the perceived preferences into their own stated preference (Q35-Q37).

Participants frequently commented on how (not) being a parent or grandparent influenced their health-state preferences. Participants who had a (grand)child were seemingly more ambivalent about their preferences than those who had not. They considered how impaired health of the one would probably affect the other (eg, they would be unhappy if their child was unhappy and vice versa) and tried to compromise between their preferences and those of their (grand)child. Participants who did not have a child anticipated that their preferences would be different if they had had a child (Q38).

Participants' preferences were further influenced by life events that would likely or could potentially take place and by personal goals that could (still) be reached. Important life events and goals were finishing primary school for a 10-year-old child, having a first love for a 15-year-old adolescent, and making memories with loved ones for an adult. Participants were less inclined to give up life-years before and more inclined to give up life-years after the event had taken place or the goal had been reached (Q39). When participants believed their life expectancy or that of a 10-year-old child or 15-year-old adolescent to be too short to reach a certain goal they challenged the assumptions underlying the valuation tasks by rephrasing health states. Many participants had the idea that their loved ones would be able to reduce their suffering and

improve their HRQOL (Q40; Q41) or that some new health technology would become available that could extend their lives (Q42; Q43), despite instructions to imagine that health states would not change over time.¹⁹

Theme E: Stating a Preference

Participants considered it relatively difficult to state health-state preferences for a 10-year-old child and 15-year-old adolescent. They particularly struggled with their feeling of inflicting health impairments upon a 10-year-old child and 15-year-old adolescent. When stating preferences for themselves, participants imagined themselves living in health states. Nevertheless, when stating preferences for a 10-year-old child and 15-year-old adolescent, they often imagined subjecting the 10-year-old child and 15-year-old adolescent to health states (Q44; Q45). Participants further struggled with the responsibility of stating preferences, particularly with giving up life-years, for a 10-year-old child and, to a lesser extent, 15-year-old adolescent. Most believed that a 10-year-old child would not yet be able to make decisions about their own (prospective) health and would, therefore, be wholly dependent on their choices (Q46; Q47). Participants believed that a 15-year-old adolescent would be capable and, therefore, should have the possibility to make such decisions for themselves (Q48). Some believed that a 10-year-old child should at least be consulted or involved in these types of decisions (Q49; Q50).

Participants did not always feel entitled to state preferences for a 10-year-old child and 15-year-old adolescent. Particularly, when they had no experience living in a health state, they felt the choices were not theirs to make (Q51). They were afraid of underestimating problems and felt uncomfortable not knowing the preferences of the 10-year-old child and 15-year-old adolescent. Some anticipated what their preferences might be to compromise between their own preferences and those of the 10-year-old child and 15-year-old adolescent (Q52). Participants were also afraid of making the wrong choice and of the negative impact this might have on their relationship with the 10-year-old child and 15-year-old adolescent. They mentioned giving up relatively few life-years for a 10-year-old child and 15-year-old adolescent, not only for getting the most out of their lives but also for fear of being considered a bad parent or being held accountable for their choice by others. Participants particularly feared that others would bear a grudge against them and that they would later come to regret their choice (Q53; Q54).

Some participants who were unwilling to give up (m)any life-years for a 10-year-old child and 15-year-old adolescent mentioned feeling selfish. Most notably when they expected the 10-year-old child and 15-year-old adolescent to suffer and to prefer giving up more life-years themselves. Participants usually let their own preferences prevail, despite knowing that this was more in their own interest than in the interest of the 10-year-old child and 15-year-old adolescent (Q55; Q56).

Discussion

The results of this study suggest that when participants value EQ-5D-Y-3L health states for themselves, their thoughts and beliefs are largely different from when they value similar health states for a 10-year-old child. The results further suggest that participants' health-state preferences for a 10-year-old child may be different from such preferences for a 15-year-old adolescent. We developed 28 categories grouped under 5 themes that illustrated the main differences that emerged from the interviews. Most notably, the themes illustrated that participants (A) deemed it more mentally and emotionally straining to perform valuation tasks for a 10-year-old child and 15-year-old adolescent than for

themselves, (B) had a stable self-image, but varied greatly in whom they imagined as 10-year-old child and 15-year-old adolescent, (C) focused on different dimensions and levels for a 15-year-old adolescent than for a 10-year-old child and themselves, (D) had various thoughts about nonhealth-related factors that influenced their preferences, and (E) gave up relatively few life-years for a 10-year-old child and 15-year-old adolescent, also to avoid others bearing a grudge against them.

To the best of our knowledge, this study is one of the first to examine why adults value EQ-5D-Y-3L health states differently for themselves than for a 10-year-old child and differently for a 10-year-old child than for a child or adolescent of a different age. As such, we are limited in our ability to compare our findings with those of others. Nonetheless, we can compare our findings with those of 2 recent studies; Dewilde et al¹³ used a mixed-methods approach and Powell et al¹⁷ used focus groups to examine why adults value similar health states differently for children than for adults. Although we used different designs and perspectives (ie, Dewilde et al¹³ examined participants' health-state preferences for a 40-year-old adult and 8-year-old child and Powell et al¹⁷ for participants themselves, themselves as a 10-year-old child, another adult, and another 10-year-old child), our findings are largely consistent.

The main strength of this study lies in the standardized interview protocol based on the EQ-5D-Y-3L valuation protocol,⁷ interviewer instructions for EQ-5D-Y valuation studies using EQ-PVT,¹⁹ and criteria for cTTO protocol compliance.²⁰ Furthermore, participants first performed the valuation tasks for themselves familiarizing themselves with the health states, valuation tasks, and thinking aloud before performing the tasks, in random order, for a 10-year-old child and 15-year-old adolescent. This strength inevitably comes with the limitation that we randomized only 2 of 3 perspectives, which may have influenced our results. For example, participants may more frequently have used heuristics when performing the tasks for themselves, had they also performed the tasks from this perspective last. Therefore, further research into the role of heuristic valuation processes is warranted. Other limitations concern the use of a think-aloud protocol for meeting the research objectives and the influence this may have had on the quality of our data. For example, participants may not have verbalized all their thoughts or have given their choices more thought than they would have under different circumstances. Furthermore, the results of this study do not provide insight into the degree to which these thoughts influence the value of these health states. Therefore, further research into the direct relationship between the thoughts of adults and the value of EQ-5D-Y-3L health states is warranted. A final limitation concerns the external validity of our findings. We conducted this study in The Netherlands, and therefore, the generalizability of our findings to other populations and cultures may be limited. Indeed, members of the public in The Netherlands may, on average, be more accustomed to and comfortable with discussing difficult and controversial issues (eg, the death of a child) than in some other countries.²⁶

Despite these limitations, our results contribute to a better understanding of why similar health states are valued differently for adults than for children and why health-state preferences for a 10-year-old child may not be representative of such preferences for the full EQ-5D-Y-3L age range. Our results have broad implications for the sample, perspective, and methods used for valuation of EQ-5D-Y-3L health states, some of which are highlighted below.

Participants generally struggled with the responsibility of stating health-state preferences for a 10-year-old child and 15-year-old adolescent. Some anticipated what the child and

adolescent's preferences might be to compromise between their own preferences and those of the 10-year-old child and 15-year-old adolescent. Participants further believed that a 15-year-old adolescent would be able and hence should have the possibility to make decisions about their (prospective) health themselves and that a 10-year-old child should at least be consulted or involved in these types of decisions. It is currently unclear whether the value of EQ-5D-Y-3L health states should solely represent adults' health-state preferences for a 10-year-old child or whether such a compromise between adults' preferences and those of members of the population for which the instrument is recommended is considered desirable. In case of the former, our results may imply that adults' preferences need to be adjusted to remove any anticipated preferences of a 10-year-old child (or child or adolescent of a different age), whereas in case of the latter, our results imply that current methods need to be further developed for eliciting and integrating such a combination of adult and child preferences.

Participants furthermore struggled with the feeling of inflicting health impairments upon a 10-year-old child and 15-year-old adolescent. Most participants were unwilling to give up (m)any life-years for a 10-year-old child and 15-year-old adolescent—even when they expected them to suffer and to prefer giving up more life-years themselves. Some participants mentioned feeling selfish for this, whereas others mentioned giving up relatively few life-years out of fear that others would bear a grudge against them or that their relationship with the 10-year-old child and 15-year-old adolescent would be negatively affected. It may be considered important to take adults' unwillingness to give up (m)any life-years for a 10-year-old child or 15-year-old adolescent into account in the value of EQ-5D-Y-3L health states. Nevertheless, it is currently unclear whether it is considered desirable that any unwillingness to give up life-years not directly related to the health states under valuation should be incorporated in value sets. In the case that this is not desirable, our results may imply that further insight into the underlying causal mechanism and into the range, intensity, and source (eg, adults' experience, personality, or risk attitude) of such unwillingness is necessary. This insight can aid in designing strategies for controlling for the possible influence of such an unwillingness on the value of EQ-5D-Y-3L health states.

Participants varied greatly in whom they imagined when stating health-state preferences for a 10-year-old child and 15-year-old adolescent. They were often ambiguous and alternated between imagining their (grand)child, themselves as a 10-year-old child or 15-year-old adolescent, and an unknown 10-year-old child or 15-year-old adolescent. Consequently, they also varied in their perceived relationship with the 10-year-old child and 15-year-old adolescent. It is currently unclear whether and, if so, of which larger population adults' health-state preferences for a 10-year-old child should be representative. For example, should they be representative of the health-state preferences of the adult general population or for the population of children and adolescents aged 8 to 15 years for whom the EQ-5D-Y-3L is recommended? In case of the former, further insight into the characteristics of adults that affect their health-state preferences for a 10-year-old child is necessary to account for these characteristics in the sampling of respondents. In case of the latter, further insight into the characteristics of children and adolescents aged 8 to 15 years that affect adults' health-state preferences for a 10-year-old child is necessary, for example, to specify these characteristics in the valuation tasks.

Differences in adults' health-state preferences for themselves, a 10-year-old child, and 15-year-old child largely resulted from differences in thoughts about nonhealth-related factors and were influenced by differences in heuristic valuation processes. These

BOX 2. Framework method.

The data were analyzed using the Framework Method in 5 stages^{22,23}:

- In stage 1, the audio recordings were transcribed verbatim by subcontractors and assessed for quality and imported into ATLAS.ti version 9 by the first author.
- In stage 2, the thematic framework was developed using an iterative approach. At the start, the first 2 authors developed a working framework based on (the comparison of) their independent coding of excerpts from the same 3 transcripts that were randomly selected by the first author. The 4 authors discussed this framework and compared it with the independent coding of excerpts from 3 randomly selected transcripts by the third and fourth author. The first author further developed the framework based on the coding of excerpts in 3 rounds of 7 or 8 transcripts. In each round, (new) codes were compared with the independent coding of 3 randomly selected transcripts by the other authors, clustered into categories and overarching themes, and gradually refined. The coding of the other authors served to validate the framework and was, therefore, less detailed than that of the first author. The framework was finalized when all transcripts were coded, and the authors agreed that new codes no longer emerged from the data.
- In stage 3, the framework was used to systematically index the transcripts.
- In stage 4, a matrix with all categories and perspectives was constructed to chart the data. This was done by entering references to relevant quotations in each cell.
- In stage 5, we translated the quotations from Dutch to English and interpreted the data making use of descriptive and explanatory accounts. These accounts provided us with insight into the range and diversity of the categories and themes and with explanations for why participants valued health states differently for themselves than for a 10-year-old child and why their health-state preferences differed between a 10-year-old child and 15-year-old adolescent.

results imply that differences between the value of adult and child health states observed in comparative and EQ-5D-Y-3L valuation studies¹⁰⁻¹⁴ may not solely result from systematic differences in the value of adult and child health states. Further insight into the ways in and the extent to which such thoughts and heuristics affect the value of adult and child health states is necessary to develop methods for controlling for such influences in valuation studies.

We hope that the results of this study will stimulate discussions and provide ideas for further—empirical and normative—research into the measurement and valuation of child health states in economic evaluations of new health technologies.

Conclusions

The results of this study indicate that differences in adults' health-state preferences for themselves and a 10-year-old child largely result from differences in thoughts about nonhealth-related factors. They further indicate that health-state preferences for a 10-year-old child may not be representative of such preferences for the full EQ-5D-Y-3L age range. Our results have broad implications and may further discussions on whether the perspective and methods currently used for measurement and valuation of child health states can be considered empirically appropriate and normatively desirable.

Supplemental Materials

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.jval.2021.12.014>.

Article and Author Information

Accepted for Publication: December 20, 2021

Published Online: February 12, 2022

doi: <https://doi.org/10.1016/j.jval.2021.12.014>

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Administrative, technical, or logistic support: Reckers-Droog

Conflict of Interest Disclosures: All authors reported receiving grants from EuroQol during the conduct of the study. Dr Karimi reported receiving nonfinancial support from the EuroQol Group outside the submitted work. Dr Verstraete reported being a member of the EuroQol Group. No other disclosures were reported.

Funding/Support: This study was funded by the EuroQol Research Foundation in The Netherlands (EQ Project 20190920).

Role of Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Ethical statement: This study was approved by the internal review board of the Erasmus School of Health Policy & Management, Erasmus University Rotterdam (reference institutional review board 20-02 Reckers-Droog, February 21, 2020).

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